Appendix A v1.5: Positivity-Preserving Phase-Space Confinement, Compact Trapping Box Theorem, and Conditional Holographic Closure

Supplementary Artifact v1.5 (Upgraded & VRK-Verified Edition) — Additive to *Physics v1.0* and Appendices A v1.1–v1.4.1

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Abstract

Appendices A v1.1 through v1.4.1 established that the three-variable phenomenological dynamical system of *Physics v1.0* admits a unique interior equilibrium whose sole linear instability is a subcritical Hopf bifurcation ($l_1 > 0$) across the interior positive parameter orthant, and demonstrated that post-Hopf trajectories ($e > e_c$) escape the physical positive orthant $\mathbb{R}^3_+$ in finite time. Implementing Option 2 of Appendix A v1.1 under the epistemic type discipline of the IntentSeal Verifiable Research Kernel (VRK v0.3.1, 10.5281/zenodo.23018459), this v1.5 supplement introduces a modified, smooth (C^∞) soft-wall dynamical system that eliminates finite-time escape while converging exponentially to the v1.0–v1.4 system in the interior as the boundary regularization widths $\epsilon_H, \epsilon_I \to 0^+$. First, we state a *conditional* holographic dictionary for Gate 0 (`CONDITIONAL_FRONTIER`) via the covariant Hubeny–Rangamani–Takayanagi (HRT) and Bousso causal-diamond bounds; for general non-geometric Hilbert-space factorizations, Gate 0 remains `OPEN`. Second, by gating the irreversible-erasure sink with $\Psi_{\epsilon_H}(H_{\rm obs}) = \tanh(H_{\rm obs}/\epsilon_H)$ and the derivative-feedback channel with $\Phi_{\epsilon_I}(\dot{I}) = \tanh(\dot{I}/\epsilon_I)$, we prove via Nagumo's subtangentiality theorem that both $\mathbb{R}^3_+$ and the gain-dependent compact box $\mathcal{B}(e) = [0, \gamma_{\max}] \times [0, H_{\max}] \times [0, \dot{I}_{\max}(e)]$ are globally forward-invariant for all $t \geq 0$ (`ANALYTIC_THEOREM`), while multi-initial-condition Poincaré return-map, cross-solver (`LSODA`/`DOP853`), Liouville orbital divergence ($\int_0^T \operatorname{div} F\, dt = -10405.41 \implies \det(DP) \approx 10^{-4519} \ll 1$), and transverse Floquet analyses ($\rho_{\rm FD}(DP) < 10^{-8}$ below finite-difference resolution at rtol $= 10^{-12}, \Delta h = 10^{-2}$) demonstrate numerically (`REPRODUCED_RESULT`) that post-Hopf trajectories settle onto an attracting periodic relaxation orbit inside $\mathcal{B}(e)$. Third, because $dH_{\rm obs}/dt|_* = 0$ at steady state, the first derivative of $\Phi_{\epsilon_I}(\dot{I})$ vanishes identically in the equilibrium Jacobian, leaving $(\gamma^*, \dot{I}^*)$ exact (`ANALYTIC_THEOREM`) and preserving $H^*_{\rm obs}$, J, e_c, and l_1 up to ϵ-dependent exponential corrections scaling as $1 - \tanh(\dot{I}^*/\epsilon_I) \sim 2e^{-2\dot{I}^*/\epsilon_I}$ and $\operatorname{sech}^2(H^*_{\rm obs}/\epsilon_H) \sim 4e^{-2H^*_{\rm obs}/\epsilon_H}$. Across $\epsilon_H = \epsilon_I = \epsilon \in [0.005, 0.10]$ on the v1.4 audit slice (`REPRODUCED_RESULT`), $\|J_{\rm v1.5} - J_{\rm v1.4}\|_F$ and $|\Delta l_1|$ decay from 2.21×10^{-8} and 1.75×10^{-6} at $\epsilon = 0.10$, to 1.48×10^{-86} and 1.11×10^{-82} at $\epsilon = 0.01$ ($\operatorname{sech}^2(H^*_{\rm obs}/\epsilon_H) = 5.38 \times 10^{-695}$), and 2.05×10^{-173} and 6.11×10^{-169} at $\epsilon = 0.005$. All six face conditions of $\mathcal{B}(e)$ are formalized in Lean 4 (`BoundaryVerification.lean`, pinned via `lakefile.lean` and `lake-manifest.json` to `leanprover/lean4:v4.36.0-rc1` and Mathlib `10156080d0e7`), and all symbolic equilibria, Taylor tensors, ϵ-scaling sweeps, Liouville/Floquet bounds, and VRK v0.3.1 claim schemas are verified via `SymPy`, 60/750/1450-digit `mpmath`, and `SciPy` (`verification_patch_v1_5.py`).

1 Lineage Provenance and VRK Epistemic Scope

Under the immutability and additive-provenance rules of the *Physics v1.0* and IntentSeal research lineages ([10.17605/OSF.IO/N3DQU], 10.5281/zenodo.23018459, 10.5281/zenodo.23016445, 10.5281/zenodo.23474505), the original unregularized ODE ansatz analyzed in Appendix A v1.1 and the exact symbolic positivity certificate of Appendix A v1.4/v1.4.1 (10.17605/OSF.IO/U7M6H, 10.17605/OSF.IO/S7WHX) remain frozen and unmodified. Specifically, Appendix A v1.1 identified two minimal modelling options to address the finite-time escape to $\gamma < 0$ across the Hopf boundary $e = e_c$:

1. Restrict the admissible parameter domain of the v1.0 ODE to the linearly stable regime $e < e_c$ with the erasure admissibility bound $\dot{I}_{\rm erase} \leq c\gamma^* H_{\max}$ enforced; or
2. Define a modified, positivity-preserving dynamical system (v1.5) in which $\mathbb{R}^3_+ = \{(\gamma, H_{\rm obs}, \dot{I}) \in \mathbb{R}^3 \mid \gamma \geq 0,\ H_{\rm obs} \geq 0,\ \dot{I} \geq 0\}$ is globally forward-invariant while converging exponentially to the v1.0–v1.4 equilibrium and Hopf scaling in the interior regime ($\epsilon_H, \epsilon_I \ll 1$).

This document executes Option 2 as a distinct versioned dynamical model (v1.5) and enforces VRK v0.3.1 epistemic type separation (`ANALYTIC_THEOREM` vs. `REPRODUCED_RESULT` vs. `CONDITIONAL_FRONTIER`/`OPEN`) across all claims.

2 Conditional Holographic Dictionary for Gate 0

In *Physics v1.0*, the observer von Neumann entropy $H_{\rm obs}(t) = -\operatorname{Tr}(\rho_{\rm obs}\log_2\rho_{\rm obs})$ is bounded by an informational saturation capacity $H_{\max}$ (in bits). The Gate 0 Technical Report recorded the bridge between an abstract subsystem factorization $\mathcal{H}_{\rm total} = \mathcal{H}_{\rm obs}\otimes\mathcal{H}_{\rm env}$ and geometric horizon entropy $A/(4\ell_P^2\ln 2)$ as **UNRESOLVED IN GENERALITY** because an arbitrary algebraic tensor factorization need not correspond to a spatially localized region with a well-defined boundary area.

To make the microphysical assumption required to close Gate 0 mathematically explicit, we formulate a conditional covariant causal-diamond localization postulate:

Proposition 1 (Conditional Holographic Closure of Gate 0)**.** *Let $(\mathcal{M}, g_{\mu\nu})$ be a $(3+1)$-dimensional semiclassical spacetime, and suppose the observer subsystem $\mathcal{H}_{obs}$ is defined by the algebra of quantum fields localized within the causal diamond $\mathcal{D}(\mathcal{A})$ of a spacelike ball $\mathcal{A}$ with smooth codimension-two boundary $\partial\mathcal{A}$. Under the covariant Bousso entropy bound and the Hubeny–Rangamani–Takayanagi (HRT) extremal surface prescription, if Σ_{HRT} is the unique minimal-area extremal surface homologous to $\mathcal{A}$ with $\partial\Sigma_{HRT} = \partial\mathcal{A}$ and stationary area $\mathrm{Area}(\Sigma_{HRT}) = A_\Sigma$, then the state-space supremum of the observer's fine-grained von Neumann entropy in bits is fixed by*

$$H_{\max} \equiv \sup_{\rho_{obs}\in\mathcal{S}(\mathcal{D}(\mathcal{A}))} [-\operatorname{Tr}(\rho_{obs}\log_2\rho_{obs})] = \frac{\mathrm{Area}(\Sigma_{HRT})}{4\ell_P^2\ln 2} \quad [bits]. \tag{1}$$

Remark 2 (Distinction Between Conditional Closure and General Derivation)*.* Equation (1) closes Gate 0 *conditionally* (`CONDITIONAL_FRONTIER`) within the class of spatially localized causal-diamond subsystems admitting a semiclassical holographic or covariant entropy bound. For abstract, non-geometric Hilbert-space factorizations $\mathcal{H}_{\rm total} = \mathcal{H}_{\rm obs}\otimes\mathcal{H}_{\rm env}$ without a spatial boundary $\partial\mathcal{A}$, Gate 0 remains `OPEN` in generality. None of the dynamical-sector theorems in Sections 3–6 depend on Eq. (1); they hold for any constant $H_{\max} > 0$.

3 Upgraded Soft-Wall Dynamical System and Global Trapping Box Theorem

In the unregularized v1.0 system, finite-time escape from $\mathbb{R}^3_+$ occurs through a two-stage boundary breach: (i) when $H_{\rm obs}$ approaches 0, the constant erasure term $-\dot{I}_{\rm erase}$ remains active and drives $H_{\rm obs} < 0$; and (ii) during large-amplitude oscillations above the subcritical Hopf threshold ($e > e_c$), negative entropy rates $dH_{\rm obs}/dt < 0$ enter the derivative-feedback term $e\,(dH_{\rm obs}/dt)$ unattenuated at $\dot{I} = 0$, driving $\dot{I} < 0$ and subsequently pulling $\gamma < 0$ via $a\dot{I}\mathcal{E}_{\rm avail}(1-\gamma/\gamma_{\max})$.

Note that replacing $-\dot{I}_{\rm erase}$ with $-\frac{1}{2}\dot{I}_{\rm erase}[1+\tanh(H_{\rm obs}/\epsilon_H)]$ or leaving $e\,(dH_{\rm obs}/dt)$ ungated at $\dot{I} = 0$ does *not* render $\mathbb{R}^3_+$ forward-invariant, because $\frac{1}{2}[1+\tanh(0)] = \frac{1}{2} \neq 0$ leaves a negative sink $-\frac{1}{2}\dot{I}_{\rm erase}$ at $H_{\rm obs} = 0$ whenever $\gamma < \dot{I}_{\rm erase}/(2cH_{\max})$, and an ungated $e\,(dH_{\rm obs}/dt)$ drives $\left.\frac{d\dot{I}}{dt}\right|_{\dot{I}=0} < 0$ whenever $e\,(dH_{\rm obs}/dt) < -f\mathcal{P}_{\rm input}$.

To construct a modified C^∞ dynamical system in which $\mathbb{R}^3_+$ and a compact box $\mathcal{B}(e)$ are unconditionally forward-invariant, let $\epsilon_H, \epsilon_I > 0$ denote boundary regularization widths ($\epsilon_H \ll H_{\max}$, $\epsilon_I \ll \dot{I}^*$) and define the smooth gates

$$\Psi_{\epsilon_H}(H_{\rm obs}) \equiv \tanh\!\left(\frac{H_{\rm obs}}{\epsilon_H}\right), \qquad \Phi_{\epsilon_I}(\dot{I}) \equiv \tanh\!\left(\frac{\dot{I}}{\epsilon_I}\right). \tag{2}$$

The v1.5 soft-wall dynamical system is defined by

$$\frac{d\gamma}{dt} = a\dot{I}\mathcal{E}_{\rm avail}\left(1-\frac{\gamma}{\gamma_{\max}}\right) - b\gamma, \tag{3}$$

$$\frac{dH_{\rm obs}}{dt} = c\gamma(H_{\max}-H_{\rm obs}) - \dot{I}_{\rm erase}\tanh\!\left(\frac{H_{\rm obs}}{\epsilon_H}\right), \tag{4}$$

$$\frac{d\dot{I}}{dt} = e\tanh\!\left(\frac{\dot{I}}{\epsilon_I}\right)\left(\frac{dH_{\rm obs}}{dt}\right) + f\mathcal{P}_{\rm input} - g\dot{I}, \tag{5}$$

where all parameters $a, b, c, e, f, g, \mathcal{E}_{\rm avail}, \mathcal{P}_{\rm input}, \dot{I}_{\rm erase}, \gamma_{\max}, H_{\max}, \epsilon_H, \epsilon_I$ are strictly positive.

Theorem 3 (Global Forward-Invariance of $\mathbb{R}^3_+$ and Gain-Dependent Compact Trapping Box $\mathcal{B}(e)$)**.** *For all strictly positive parameters and every feedback gain $e > 0$:*

1. *The non-negative orthant* $\mathbb{R}^3_+ = \{(\gamma, H_{obs}, \dot{I}) \in \mathbb{R}^3 \mid \gamma \geq 0,\, H_{obs} \geq 0,\, \dot{I} \geq 0\}$ *is positively invariant under the flow of Eqs.* (3)–(5).
2. *Define the maximum entropy-rate bound* $M_H \equiv c\gamma_{\max} H_{\max} + \dot{I}_{erase}$ *and the* e*-dependent processing ceiling*

$$\dot{I}_{\max}(e) \equiv \frac{eM_H + f\mathcal{P}_{input}}{g} > 0, \tag{6}$$

which scales affinely with the feedback gain e*. Then the compact box* $\mathcal{B}(e) \equiv [0, \gamma_{\max}] \times [0, H_{\max}] \times [0, \dot{I}_{\max}(e)] \subset \mathbb{R}^3_+$ *is positively invariant and globally absorbing within* $[0, \gamma_{\max}] \times [0, H_{\max}] \times [0, \infty)$*. Consequently, every trajectory originating in* $\mathcal{B}(e)$ *exists globally for all* $t \in [0, \infty)$ *and never escapes to negative states or blows up in finite time.*

Proof. Since the vector field $F = (d\gamma/dt, dH_{\rm obs}/dt, d\dot{I}/dt)$ of Eqs. (3)–(5) is C^∞ on $\mathbb{R}^3$, by Nagumo's subtangentiality theorem a closed convex box $\mathcal{B}(e) = \prod_{i=1}^3 [x_{i,\min}, x_{i,\max}]$ is positively invariant if and only if $F_i(x) \geq 0$ whenever $x \in \mathcal{B}(e)$ with $x_i = x_{i,\min}$ and $F_i(x) \leq 0$ whenever $x \in \mathcal{B}(e)$ with $x_i = x_{i,\max}$. We check all six boundary faces of $\mathcal{B}(e)$:

1. **Face 1 — Lower γ-face ($\gamma = 0$, $\dot{I} \geq 0$):** Evaluating Eq. (3) at $\gamma = 0$ yields

$$\left.\frac{d\gamma}{dt}\right|_{\gamma=0} = a\dot{I}\mathcal{E}_{\rm avail}(1 - 0) - b(0) = a\dot{I}\mathcal{E}_{\rm avail} \geq 0, \tag{7}$$

   with strict positivity whenever $\dot{I} > 0$.

2. **Face 2 — Upper γ-ceiling ($\gamma = \gamma_{\max}$, $\dot{I} \geq 0$):** Evaluating Eq. (3) at $\gamma = \gamma_{\max}$ yields

$$\left.\frac{d\gamma}{dt}\right|_{\gamma=\gamma_{\max}} = a\dot{I}\mathcal{E}_{\rm avail}\left(1 - \frac{\gamma_{\max}}{\gamma_{\max}}\right) - b\gamma_{\max} = -b\gamma_{\max} < 0. \tag{8}$$

3. **Face 3 — Lower $H_{\rm obs}$-face ($H_{\rm obs} = 0$, $\gamma \geq 0$):** Because $\tanh(0/\epsilon_H) = \tanh(0) = 0$, evaluating Eq. (4) at $H_{\rm obs} = 0$ gives

$$\left.\frac{dH_{\rm obs}}{dt}\right|_{H_{\rm obs}=0} = c\gamma(H_{\max} - 0) - \dot{I}_{\rm erase}\tanh(0) = c\gamma H_{\max} \geq 0 \quad \forall\gamma \geq 0. \tag{9}$$

4. **Face 4 — Upper $H_{\rm obs}$-ceiling ($H_{\rm obs} = H_{\max}$, $\gamma \geq 0$):** Since $H_{\max}/\epsilon_H > 0 \implies \tanh(H_{\max}/\epsilon_H) > 0$,

$$\left.\frac{dH_{\rm obs}}{dt}\right|_{H_{\rm obs}=H_{\max}} = c\gamma(0) - \dot{I}_{\rm erase}\tanh\left(\frac{H_{\max}}{\epsilon_H}\right) < 0. \tag{10}$$

5. **Face 5 — Lower $\dot{I}$-face ($\dot{I} = 0$, arbitrary $\gamma, H_{\rm obs} \in \mathbb{R}$):** Because $\tanh(0/\epsilon_I) = 0$, both the feedback term $e\tanh(0)(dH_{\rm obs}/dt)$ and the linear damping $-g(0)$ vanish identically at $\dot{I} = 0$:

$$\left.\frac{d\dot{I}}{dt}\right|_{\dot{I}=0} = e \cdot 0 \cdot \frac{dH_{\rm obs}}{dt} + f\mathcal{P}_{\rm input} - g(0) = f\mathcal{P}_{\rm input} > 0. \tag{11}$$

6. **Face 6 — Upper $\dot{I}$-ceiling ($\dot{I} = \dot{I}_{\max}(e)$ on $\mathcal{B}(e)$):** For any $(\gamma, H_{\rm obs}) \in [0, \gamma_{\max}] \times [0, H_{\max}]$, $dH_{\rm obs}/dt \leq c\gamma_{\max}H_{\max} \leq c\gamma_{\max}H_{\max} + \dot{I}_{\rm erase} = M_H$. Since $0 \leq \tanh(\dot{I}/\epsilon_I) \leq 1$, at $\dot{I} = \dot{I}_{\max}(e)$ we have

$$\left.\frac{d\dot{I}}{dt}\right|_{\dot{I}=\dot{I}_{\max}(e)} \leq eM_H + f\mathcal{P}_{\rm input} - g\dot{I}_{\max}(e) = 0. \tag{12}$$

Since F points inward on every boundary face of $\mathcal{B}(e)$ and $\mathcal{B}(e)$ is compact, standard ODE global existence guarantees that all trajectories in $\mathcal{B}(e)$ are defined and confined to $\mathcal{B}(e)$ for all $t \geq 0$. □

VRK Type Separation — Analytical Containment vs. Numerical Relaxation Limit Cycle: Under VRK v0.3.1 (10.5281/zenodo.23018459), Theorem 3 is classified as `ANALYTIC_THEOREM` (global forward-invariance of $\mathcal{B}(e)$), whereas the post-Hopf relaxation limit cycle ($e > e_c$) is classified as `REPRODUCED_RESULT`, certified across 3 initial conditions in $\mathcal{B}(e)$ and independent `LSODA`/`DOP853` integrators via Poincaré return-map integration, Liouville orbital divergence ($\int_0^T \operatorname{div} F\, dt = -10405.41 \implies \det(DP) \approx 10^{-4519} \ll 1$), and finite-difference transverse Floquet bounds ($\rho_{\rm FD}(DP) < 10^{-8}$ below finite-difference resolution at rtol $= 10^{-12}$, $\Delta h = 10^{-2}$, Section 6, Table 3).

4 Formal Kernel Verification in Lean 4 (All Six Faces of $\mathcal{B}(e)$)

The accompanying Lean 4 Lake module `BoundaryVerification.lean` (built via `lake build` under `leanprover/lean4:v4.36.0-rc1` and Mathlib `10156080d0e7`) formalizes Eqs. (3)–(6) over Mathlib's $\mathbb{R}$ and proves all six boundary face conditions, the auxiliary bound `dHobs_dt_le_M_H`, and the composite 6-face certificate `compact_box_nagumo_certificate` using explicit Mathlib monotonicity lemmas (`mul_le_mul_of_nonneg_left`, `mul_le_mul_of_nonneg_right`, `mul_assoc`, `mul_div_cancel₀`, and `linarith`) with zero `sorry` axioms and zero `nlinarith` heuristics:

```
/-- Upper ceiling theorems (Faces 2, 4, 6) and lower face theorems (Faces 1, 3, 5). -/
theorem dHobs_dt_neg_at_ceiling
    (c H_max I_erase epsH : ℝ) (hIe : 0 < I_erase) (htanh : 0 < Real.tanh (H_max / epsH))
    (s : CosmologicalState) (hH : s.H_obs = H_max) :
    dHobs_dt c H_max I_erase epsH s < 0 := by
  unfold dHobs_dt; rw [hH, sub_self, mul_zero, zero_sub]
  exact neg_lt_zero.mpr (mul_pos hIe htanh)

theorem dIdot_dt_nonpos_at_ceiling
    (e f g P_input c gamma_max H_max I_erase epsH epsI : ℝ)
    (he : 0 ≤ e) (hg : 0 < g) (hMH : 0 ≤ M_H c gamma_max H_max I_erase)
    (s : CosmologicalState)
    (hI : s.I_dot = I_dot_max e f g P_input c gamma_max H_max I_erase)
    (hdH : dHobs_dt c H_max I_erase epsH s ≤ M_H c gamma_max H_max I_erase)
    (htanh_nonneg : 0 ≤ Real.tanh (s.I_dot / epsI))
    (htanh_le_one : Real.tanh (s.I_dot / epsI) ≤ 1) :
    dIdot_dt e f g P_input c H_max I_erase epsH epsI s ≤ 0 := by
  unfold dIdot_dt I_dot_max at *
  have hg_ne : g ≠ 0 := ne_of_gt hg
  have h_g_mul : g * s.I_dot = e * M_H c gamma_max H_max I_erase + f * P_input := by
    rw [hI]; exact mul_div_cancel₀ (e * M_H c gamma_max H_max I_erase + f * P_input) hg_ne
  have h_step1 : Real.tanh (s.I_dot / epsI) * dHobs_dt c H_max I_erase epsH s ≤
      Real.tanh (s.I_dot / epsI) * M_H c gamma_max H_max I_erase :=
    mul_le_mul_of_nonneg_left hdH htanh_nonneg
  have h_step2 : Real.tanh (s.I_dot / epsI) * M_H c gamma_max H_max I_erase ≤
      1 * M_H c gamma_max H_max I_erase :=
    mul_le_mul_of_nonneg_right htanh_le_one hMH
  have h_step3 : Real.tanh (s.I_dot / epsI) * dHobs_dt c H_max I_erase epsH s ≤
      M_H c gamma_max H_max I_erase := by linarith
  have h_step4 : e * (Real.tanh (s.I_dot / epsI) * dHobs_dt c H_max I_erase epsH s) ≤
      e * M_H c gamma_max H_max I_erase :=
    mul_le_mul_of_nonneg_left h_step3 he
  have h_assoc : e * Real.tanh (s.I_dot / epsI) * dHobs_dt c H_max I_erase epsH s =
      e * (Real.tanh (s.I_dot / epsI) * dHobs_dt c H_max I_erase epsH s) :=
    mul_assoc e (Real.tanh (s.I_dot / epsI)) (dHobs_dt c H_max I_erase epsH s)
  linarith
```

5 Exactness of $(\gamma^*, \dot{I}^*)$ and Multi-Scale Gate Perturbation Bounds

Gating only the erasure sink $\dot{I}_{\text{erase}} \tanh(H_{\text{obs}}/\epsilon_H)$ and the feedback channel $e \tanh(\dot{I}/\epsilon_I)(dH_{\text{obs}}/dt)$—while keeping $-b\gamma$ and $-g\dot{I}$ linear—ensures that $(\gamma^*, \dot{I}^*)$ are exact and the first-order gate derivative vanishes identically at equilibrium:

Proposition 4 (Exactness of $(\gamma^*, \dot{I}^*)$ and Multi-Scale Gate Perturbation Bounds)**.** *At any equilibrium $(\gamma^*, H^*_{obs}, \dot{I}^*)$ of Eqs. (3)–(5):*

1. *Since $dH_{obs}/dt|_* = 0$, Eq. (5) reduces at steady state to $f\mathcal{P}_{input} - g\dot{I}^* = 0$ and Eq. (3) reduces to $K(1 - \gamma^*/\gamma_{\max}) - b\gamma^* = 0$ with $K \equiv a\mathcal{E}_{avail}\dot{I}^*$. Thus $\dot{I}^* = f\mathcal{P}_{input}/g$ and $\gamma^* = K\gamma_{\max}/(b\gamma_{\max} + K)$ are **exact** (zero ϵ-dependence), while H^*_{obs} solves $c\gamma^*(H_{\max} - H^*_{obs}) = \dot{I}_{erase} \tanh(H^*_{obs}/\epsilon_H)$ and differs from $H_{\max} - \dot{I}_{erase}/(c\gamma^*)$ by $O(e^{-2H^*_{obs}/\epsilon_H})$.*
2. *When linearizing Eq. (5) about the equilibrium, the product rule applied to $e \tanh(\dot{I}/\epsilon_I)(dH_{obs}/dt)$ yields*

$$\frac{\partial}{\partial \dot{I}}\left[e \tanh\left(\frac{\dot{I}}{\epsilon_I}\right)\frac{dH_{obs}}{dt}\right]\Bigg|_* = \frac{e}{\epsilon_I}\operatorname{sech}^2\left(\frac{\dot{I}^*}{\epsilon_I}\right)\underbrace{\frac{dH_{obs}}{dt}\Big|_*}_{=0} + e \tanh\left(\frac{\dot{I}^*}{\epsilon_I}\right)\underbrace{\frac{\partial(dH_{obs}/dt)}{\partial \dot{I}}\Big|_*}_{=0} = 0. \tag{13}$$

Hence the $(3,3)$ *entry of the Jacobian is exactly* $-g$*, with no* $\operatorname{sech}^2(\dot{I}^*/\epsilon_I)$ *linear damping artifact.*

3. *In coordinates* $(\gamma, H_{obs}, \dot{I})$*, the exact Jacobian of the upgraded system* (3)–(5) *at equilibrium is*

$$J_{v1.5} = \begin{pmatrix} -\alpha & 0 & \beta \\ \delta_\epsilon & -\mu_\epsilon & 0 \\ e_{eff}\delta_\epsilon & -e_{eff}\mu_\epsilon & -g \end{pmatrix}, \tag{14}$$

where α *and* β *are identical to v1.1–v1.4,* $e_{eff} = e\tanh(\dot{I}^*/\epsilon_I)$*,* $\delta_\epsilon = c(H_{\max} - H^*_{obs})$*, and* $\mu_\epsilon = c\gamma^* + \frac{\dot{I}_{erase}}{\epsilon_H}\operatorname{sech}^2(H^*_{obs}/\epsilon_H)$*. Two distinct exponential gate scales govern the departure from v1.4 as* $\epsilon_H, \epsilon_I \to 0^+$*:*

- ***Processing gate scale*** *(*ϵ_I*):* $1 - \tanh(\dot{I}^*/\epsilon_I) \sim 2e^{-2\dot{I}^*/\epsilon_I}$*, which dominates* $\|J_{v1.5} - J_{v1.4}\|_F = O(e^{-2\dot{I}^*/\epsilon_I})$ *and* $|e_{c,v1.5} - e_{c,v1.4}|$*, and via the second and third derivatives of* $\tanh(\dot{I}/\epsilon_I)$ *shifts* l_1 *by* $O(\epsilon_I^{-2}\operatorname{sech}^2(\dot{I}^*/\epsilon_I))$*. Because* $l_{1,v1.4} > 0$ *holds symbolically across the positive parameter orthant ([2],* `ANALYTIC_THEOREM`*),* $l_{1,v1.5} > 0$ *is preserved by continuity for sufficiently small* $\epsilon_H, \epsilon_I > 0$ *and verified numerically across* $\epsilon \in [0.005, 0.10]$ *(*`REPRODUCED_RESULT`*).*
- ***Entropy gate scale*** *(*ϵ_H*):* $\operatorname{sech}^2(H^*_{obs}/\epsilon_H) \sim 4e^{-2H^*_{obs}/\epsilon_H}$*, which shifts* H^*_{obs} *and* μ_ϵ *by* $O(e^{-2H^*_{obs}/\epsilon_H})$*.*

6 Symbolic Tensor Audit, ϵ-Scaling Sweep, VRK Audit, and Benchmark

The script `verification_patch_v1_5.py` executes a six-layer mechanical verification chain across two standard parameter slices from the prior lineage:

- **v1.3/v1.4 Audit Slice (Tables 1 and 2):** $a = 0.5, b = 1, c = 0.5, f = g = 1, \mathcal{E}_{\text{avail}} = 2, \mathcal{P}_{\text{input}} = 1, \dot{I}_{\text{erase}} = 0.5, \gamma_{\max} = 1, H_{\max} = 10$, where $e_c = 135/26 \approx 5.192308$ and $(\gamma^*, H^*_{\text{obs}}, \dot{I}^*) = (0.5, 8.0, 1.0)$.
- **v1.1 Dynamical Benchmark (Table 3):** $a = b = c = f = g = \mathcal{E}_{\text{avail}} = 1, \gamma_{\max} = 10, H_{\max} = 100, \mathcal{P}_{\text{input}} = 5, \dot{I}_{\text{erase}} = 10$, where $e_c = 44.880952/\dot{I}_{\text{erase}} = 4.488095$ and $(\gamma^*, H^*_{\text{obs}}, \dot{I}^*) = (10/3,\ 97.0,\ 5.0)$.
- **Six-Layer Verification Chain & VRK v0.3.1 Gate:** (1) **Lean 4 Lake Kernel** (`Boundary Verification.lean`, `v4.36.0-rc1`, Mathlib `10156080d0e7`) certifies all 6 face/ceiling conditions of $\mathcal{B}(e)$ (`compact_box_nagumo_certificate`, Scope: Theorem 3); (2) **SymPy Symbolic Audit** verifies $f_1 = f_2 = f_3 = 0$ at steady state and $J_{\text{v1.5}}, B_{\text{v1.5}}, C_{\text{v1.5}}$ ($C_1 = 0$) with zero symbolic residual (Proposition 4(1)–(2)); (3) **60-Digit `mpmath`** verifies $l_{1,\text{v1.4}}$ against the 50-digit exact reference ($|\Delta| < 10^{-51}$); (4) **750/1450-Digit `mpmath`** certifies the two-gate perturbation bounds and 5-point ϵ-scaling sweep against leading-order asymptotic formulas (Tables 1–2); (5) **`LSODA/DOP853` + 3-IC Poincaré + Liouville/Floquet Audit** certifies post-Hopf confinement, cross-solver replication ($\|y_{\text{LSODA}} - y_{\text{DOP853}}\|_2 < 10^{-6}$), 3-IC convergence ($\max \|\Delta y_{\text{IC}}\|_2 < 10^{-6}$), Liouville orbital divergence ($\int_0^T \operatorname{div} F\, dt = -10405.41 \implies \det(DP) \approx 10^{-4519} \ll 1$), and finite-difference transverse Floquet upper bounds ($\rho_{\text{FD}}(DP) < 10^{-8}$ below finite-difference resolution at rtol $= 10^{-12}, \Delta h = 10^{-2}$, Table 3); and (6) **IntentSeal VRK v0.3.1 Claim-Schema & Evidence-Manifest Gate** (10.5281/zenodo.23018459, trust root `f010a99d...05a0d0f`) validates 4 claims (`ANALYTIC_THEOREM` ×2, `REPRODUCED_RESULT` ×2) against on-disk files/anchors, rejects illegal `REPRODUCED_RESULT` → `ANALYTIC_THEOREM` promotion mutations, and produces deterministic certificate digest `5eb316f25edef8e711db7ddec88224172a9cadd8cf7d332457dfa596f4368117`.

Table 1: High-precision spectral, Frobenius-norm, two-gate scale, and VRK v0.3.1 verification at the v1.3/v1.4 audit slice ($\epsilon_H = \epsilon_I = 0.01$, $e_c = 135/26$).

Quantity	**Verified Value (60/750-Digit `mpmath`, `SymPy` & VRK v0.3.1 Gate)**
SymPy equilibrium & tensor audit	Exact zero residual for $f_1 = f_2 = f_3 = 0$ and $J_{\mathrm{v1.5}}, B_{\mathrm{v1.5}}, C_{1,\mathrm{v1.5}} = 0, C_{2,\mathrm{v1.5}}, C_{3,\mathrm{v1.5}}$
Analytic Hopf threshold e_c & ω_c	$e_c = 135/26 = 5.19230769230769\ldots$, $\omega_c = \sqrt{2/13} = 0.392232270276368\ldots$ s^{-1}
First Lyapunov coefficient $l_1 = \mathrm{Re}(G_{21})/(2\omega_c)$	$+0.519589117351220268983393 19\ldots$ (subcritical, $l_1 > 0$)
Processing gate deviation $1 - \tanh(\dot{I}^*/\epsilon_I)$	$2.76779305347 \times 10^{-87}$ ($\dot{I}^*/\epsilon_I = 100$)
Entropy gate deviation $\mathrm{sech}^2(H^*_{\mathrm{obs}}/\epsilon_H)$	$5.38132158610 \times 10^{-695}$ ($H^*_{\mathrm{obs}}/\epsilon_H = 800$)
Soft-wall equilibrium shift $\lvert H^*_{\mathrm{obs,v1.5}} - H^*_{\mathrm{obs,v1.4}}\rvert$	$5.38132158610 \times 10^{-695}$ ($< 10^{-694} \ll 10^{-50}$)
Jacobian Frobenius norm $\lVert J_{\mathrm{v1.5}} - J_{\mathrm{v1.4}}\rVert_F$	$1.48135280746 \times 10^{-86}$ ($< 10^{-85} \ll 10^{-50}$)
Hopf threshold shift $\lvert e_{c,\mathrm{v1.5}} - e_{c,\mathrm{v1.4}}\rvert$	$1.43712331623 \times 10^{-86}$ ($< 10^{-85} \ll 10^{-50}$)
First Lyapunov shift $\lvert l_{1,\mathrm{v1.5}} - l_{1,\mathrm{v1.4}}\rvert$	$1.10793084174 \times 10^{-82}$ ($< 10^{-81} \ll 10^{-50}$)
Linear gate derivative $\partial_{\dot{I}}[e\tanh(\dot{I}/\epsilon_I)\dot{H}_{\mathrm{obs}}]_*$	0 (vanishes identically since $dH_{\mathrm{obs}}/dt\rvert_* = 0$)
VRK v0.3.1 certificate digest	`5eb316f25edef8e711db7ddec88224172a9cadd8cf7d332457dfa596f4368117`

Table 2: Parametric gate-width scaling sweep on the v1.3/v1.4 audit slice ($\epsilon_H = \epsilon_I = \epsilon$), demonstrating exponential convergence of the v1.5 soft-wall system to v1.4 as $\epsilon \to 0^+$.

Gate Width ϵ	$1 - \tanh(\dot{I}^*/\epsilon)$	$\lVert J_{\mathbf{v1.5}} - J_{\mathbf{v1.4}}\rVert_F$	$\lvert e_{c,\mathbf{v1.5}} - e_{c,\mathbf{v1.4}}\rvert$	$\lvert l_{1,\mathbf{v1.5}} - l_{1,\mathbf{v1.4}}\rvert$
$\epsilon = 0.100$	4.12231×10^{-9}	2.20630×10^{-8}	2.14043×10^{-8}	1.74819×10^{-6}
$\epsilon = 0.050$	8.49671×10^{-18}	4.54753×10^{-17}	4.41175×10^{-17}	1.39640×10^{-14}
$\epsilon = 0.020$	7.44015×10^{-44}	3.98205×10^{-43}	3.86316×10^{-43}	7.49478×10^{-40}
$\epsilon = 0.010$	2.76779×10^{-87}	1.48135×10^{-86}	1.43712×10^{-86}	1.10793×10^{-82}
$\epsilon = 0.005$	3.83034×10^{-174}	2.05004×10^{-173}	1.98883×10^{-173}	6.11280×10^{-169}

Table 3: Post-Hopf trajectory confinement on the v1.1 benchmark ($\dot{I}_{\mathrm{erase}} = 10, e_c = 4.488095, \epsilon_H = \epsilon_I = 0.05$). At $e = 1.20\,e_c$, a Poincaré return map on $\Sigma = \{\gamma = 1.0\ \mathrm{s}^{-1}, \dot{\gamma} > 0\}$ replicated across `LSODA` and `DOP853` ($\lVert y_{\mathrm{LSODA}} - y_{\mathrm{DOP853}}\rVert_2 < 10^{-6}$) confirms convergence across 3 initial conditions in $\mathcal{B}(e)$ ($\max \lVert \Delta y_{\mathrm{IC}}\rVert_2 < 10^{-6}$) to an attracting periodic relaxation orbit at $(H_{\mathrm{obs}}, \dot{I})_\Sigma = (3.91083047, 21.68392735)$ with period $T = 18.957047$ s (5-cycle spread $< 10^{-6}$ s), Liouville orbital divergence integral $\int_0^T \mathrm{div}\, F\, dt = -10405.41$ ($\det(DP) \approx 10^{-4519} \ll 1$), and finite-difference transverse Floquet spectral radius $\rho_{\mathrm{FD}}(DP) < 10^{-8}$ at $\mathrm{rtol} = 10^{-12}, \Delta h = 10^{-2}$ (below the $\sim \mathrm{rtol}/(2\Delta h) \sim 10^{-6}$ noise floor at $\mathrm{rtol} = 10^{-9}, \Delta h = 10^{-3}$).

Gain	**v1.0 / v1.1 Behaviour**	**v1.5** $\min\gamma(t)$	**v1.5** $\max\gamma(t)$	**v1.5** $\min H_{\mathbf{obs}}(t)$	**v1.5 Observed Regime in** $\mathcal{B}(e)$ $(t \in [0, 400])$
$e = 0.90\,e_c$	Decays to eq. (6)	3.333272	3.333433	96.999947	Asymptotically stable equilibrium
$e = 1.01\,e_c$	Bounded osc. (5.4×10^{-3})	3.331644	3.334981	96.998534	Bounded interior oscillation
$e = 1.04\,e_c$	Escapes: $\gamma \to 0^-$ at $t \approx 259$	0.005682	9.720486	0.003339	Attracting relaxation cycle in $\mathcal{B}(e)$
$e = 1.08\,e_c$	Escapes: $\gamma \to 0^-$ at $t \approx 136$	0.005461	9.730970	0.003222	Attracting relaxation cycle in $\mathcal{B}(e)$
$e = 1.20\,e_c$	Escapes: $\gamma \to 0^-$ at $t \approx 55$	0.004889	9.758112	0.002921	Attracting cycle ($T = 18.957047$ s, $\rho_{\mathrm{FD}} < 10^{-8}$)
$e = 2.00\,e_c$	Escapes: $\gamma \to 0^-$ at $t \approx 19$	0.002885	9.854966	0.001854	Attracting relaxation cycle in $\mathcal{B}(e)$

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